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Preparation and properties of lead halide phosphate glasses [☆]

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Abstract

The glass-forming regions in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ and $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ systems were explored and glasses were prepared and properties measured. Based on these glass-forming systems, the exploration for glass-forming regions was extended to multi-component systems containing amounts of $\text{PbO} \leq 15$ mol% and/or $\text{PbF}_2 \leq 5$ mol% in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems. We show that the glass-forming regions in the ternary systems extended to 30 mol% P_2O_5 . The dissolution rate of the samples in water is in the order of 10^{-5} mm/day. These samples have densities in the range 4.5–5.81 g/cm³ and have glass transition temperatures in the range 146–281°C. The composition of the melted glasses differs from the batch composition, which is caused by the loss of some components during melting. © 2000 Elsevier Science B.V. All rights reserved.

1. Introduction

To dope organics into inorganic glasses, the melting temperature of the glasses must be low enough so that the organic will not decompose during doping. Tick [1] developed a series of lead tin phosphate glasses. These glasses had densities of up to 4.5 g/cm³ and could be melted at about 500°C. The viscosity of the glass liquids is low enough at about 300°C so that organics can be doped into the glass liquid without decomposition. A great amount of work has been done on the structure of the glasses [2–5] and on doping organics into the glasses for developing non-linear optical materials and scintillating materials [6–10].

The purpose of this work is to explore new low melting glasses with larger density in the lead halide phosphate systems. The glasses are to be used as the matrix for doping with organics.

2. Experiments

The raw materials used in the glass preparation are $\text{NH}_4\text{H}_2\text{PO}_4$, PbO , PbF_2 , PbBr_2 and PbCl_2 , among which PbBr_2 and PbCl_2 are synthesized from analytical grade lead nitrate, potassium bromide and potassium chloride through the following reactions:



The raw materials were weighed to an accuracy of ± 0.1 g (the relative error of the batch composition of a glass is calculated to be in the range 1–5%) and glasses were melted in porcelain crucibles

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in a ventilated muffle furnace. The glasses in the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems were melted at about 600°C for 15 min and those in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems were melted at $450\text{--}470^\circ\text{C}$ for 20 min. Glass samples were formed by quenching the liquids in a carbon steel mould at room temperature. The sizes of the glass samples were 20 mm in diameter and 5 mm in thickness. The glass samples were annealed at $\sim 8^\circ\text{C}$ above the glass transition temperature for 1 h and cooled in the furnace.

The quenched samples were visually classified as glasses, partially crystalline substances and crystalline substances. Differential scanning calorimetry (DSC) was used to measure the glass transition temperature (T_g) and the crystallization starting temperature (T_x). The error of the measured temperature was $\pm 2^\circ\text{C}$. The densities of the samples were measured by the Archimedes method with deionized water as the immersing medium. The relative error of the measured density was $\pm 0.2\%$. The compositions of the samples were determined by chemical analysis for F, P, Pb, Cl and Br. The oxygen content was assumed to be the remainder. The relative error of this method is normally $\pm 1\%$. The water durability of the samples was evaluated by two methods. One was to leave the glasses in air for 180 days and to observe the change of the glass surface visually, and the other was to measure the dissolution rate of the glasses in water by putting the samples into deionized water at room temperature for 72 h and measuring their weight change. The dissolution rate was cal-

culated through $R = W_L / (d\pi D(0.5D + t))$, where R is the dissolution rate, W_L the weight loss, d the density of the glass sample and D and t are the diameter and thickness of the glass sample. The relative error of the obtained dissolution rate was estimated to be $\pm 5\%$.

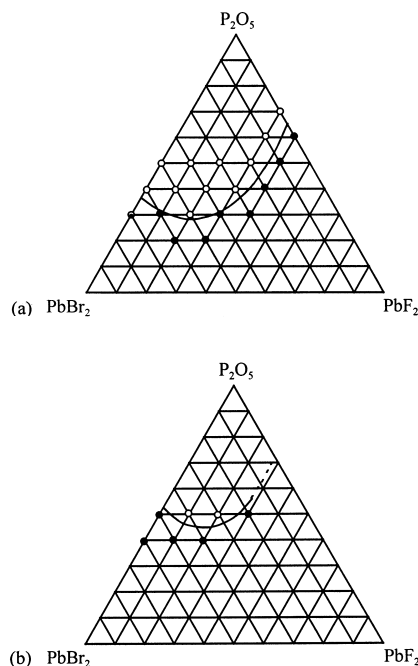


Fig. 2. Glass-forming regions in $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems (melting temperature: 600°C). (a): containing 10 mol% PbO ; (b): containing 15 mol% PbO . $\circ$: glass; $\bullet$: partially crystalline; $\bullet$: crystalline or unmelted.

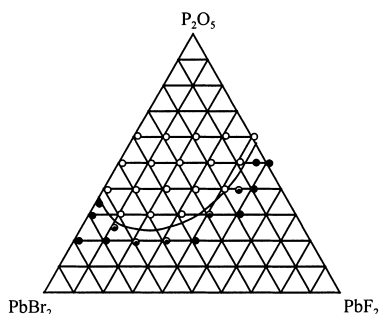


Fig. 1. Glass-forming region in $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary systems (melting temperature: 600°C). $\circ$: glass; $\bullet$: partially crystalline; $\bullet$: crystalline or unmelted.

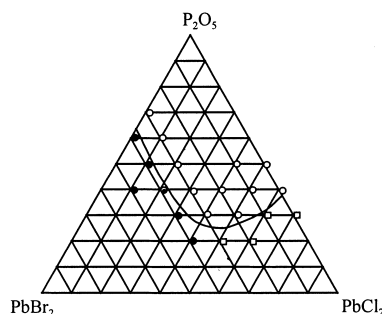


Fig. 3. Glass-forming region in $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary systems (melting temperature: $450\text{--}470^\circ\text{C}$). $\circ$: glass; $\bullet$: partially crystalline; $\bullet$: crystalline or unmelted.

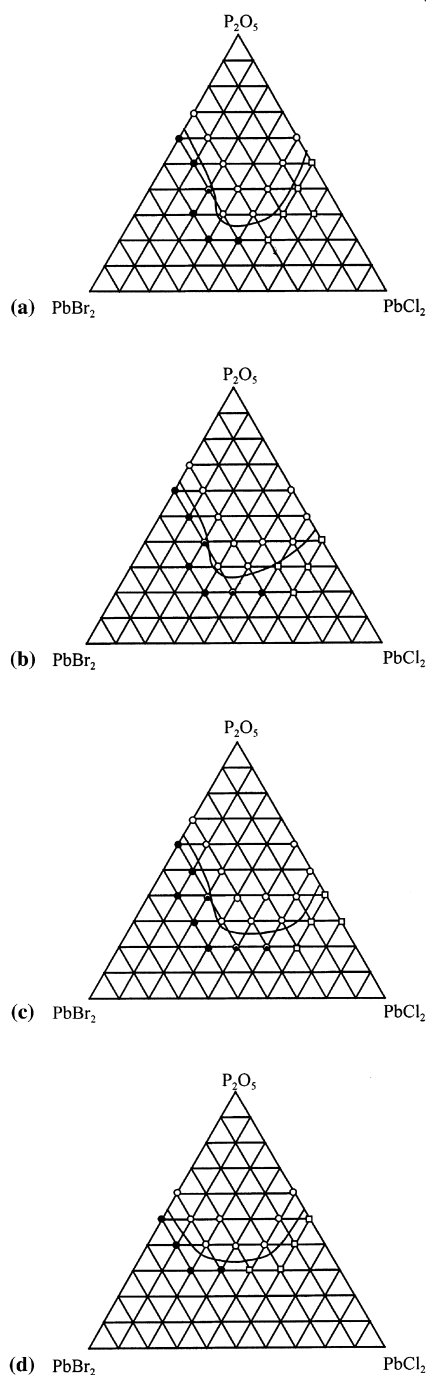


Fig. 4. Glass-forming regions in $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems (melting temperature: $450\text{--}470^\circ\text{C}$). (a): containing 5 mol% PbO ; (b): containing 5 mol% PbF_2 ; (c): containing 2.5 mol% PbO and 2.5 mol% PbF_2 ; (d) containing 5 mol% PbO and 5 mol% PbF_2 . $\circ$: glass; $\bullet$: partially crystalline; $\bullet$: crystalline or unmelted.

3. Results

Chemical analysis did not detect silicon or aluminum in the glass samples, indicating that the porcelain crucible did not contaminate the glass melts to detectable amounts.

Figs. 1–4 give the glass-forming regions in the systems of $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$, $\text{PbO--PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$, $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$, $\text{PbO--PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$, $\text{PbF}_2\text{--PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ and $\text{PbO--PbF}_2\text{--PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$. The figures indicate that there are glass-forming regions in these systems. In the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ and $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary systems, the smallest P_2O_5 content in the glass samples were 30 mol%. When 10 or 15 mol% PbO was introduced into the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary system, the glass-forming region became smaller (Fig. 2). When 5 mol% PbO or 5 mol% PbF_2 was introduced into the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary system, the glass-forming region extended towards larger PbBr_2 content and retracted towards smaller PbCl_2 content (Fig. 4(a) and (b)). Instead, if 2.5 mol% PbO and 2.5 mol% PbF_2 are introduced together, the glass-forming region is larger (Fig. 4(c)) than that shown in either Fig. 4(a) or (b). When the amount of both PbO and PbF_2 increase to 5 mol%, the glass-forming region became much smaller (Fig. 4(d)).

Fig. 5 shows the glass-forming region in the same system (the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary system) as Fig. 3, but the samples were melted at $\sim 540^\circ\text{C}$ for 15 min. The glass-forming region in

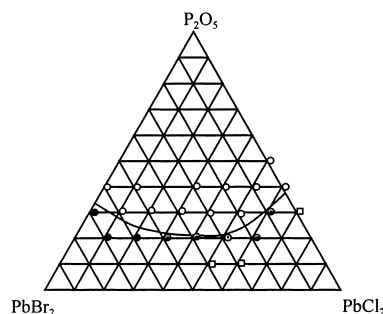


Fig. 5. Glass-forming region in $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary systems (melting temperature: $530\text{--}540^\circ\text{C}$). $\circ$: glass; $\bullet$: partially crystalline; $\bullet$: crystalline or unmelted.

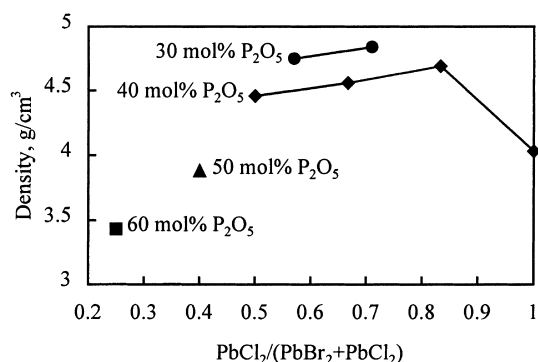


Fig. 6. Density vs. composition of glasses in the PbCl_2 – PbBr_2 – P_2O_5 ternary system. Lines are drawn as guides to the eye.

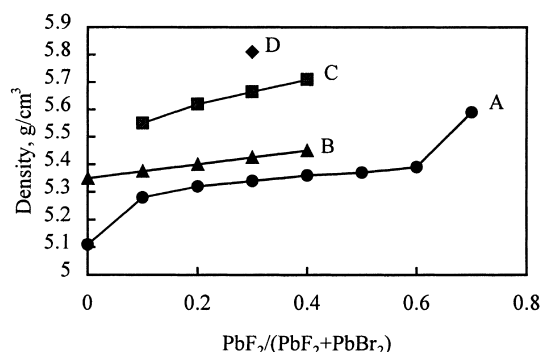


Fig. 7. Density vs. composition of glasses in the PbF_2 – PbBr_2 – P_2O_5 based systems. A: 10 mol% P_2O_5 ; B: 36 mol% P_2O_5 + 10 mol% PbO ; C: 30 mol% P_2O_5 ; D: 27 mol% P_2O_5 + 10 mol% PbO . Lines are drawn as guides to the eye.

Fig. 5 is apparently larger than that shown in Fig. 3.

Tables 1 and 2 give the batch composition and analyzed composition of some samples. The change of the composition indicates that some components were lost during melting.

The characteristic temperatures, stability indexes, and densities of some glasses are listed in Tables 3 and 4. It is shown that the glasses in the PbCl_2 – PbBr_2 – P_2O_5 based systems have lower glass transition temperatures than those in the PbF_2 – PbBr_2 – P_2O_5 based systems. The addition of 10 mol% PbO into the PbF_2 – PbBr_2 – P_2O_5 ternary system decreases the glass transition temperature while the addition of PbO and/or PbF_2 into the PbCl_2 – PbBr_2 – P_2O_5 ternary system increases the glass transition temperature. In both cases, the density of the samples are increased since the density of either PbO or PbF_2 is larger than the other components. The lowest T_g in the PbCl_2 – PbBr_2 – P_2O_5 based systems is 146°C with a density of 4.75 g/cm^3 while that in the PbF_2 – PbBr_2 – P_2O_5 based systems is 210°C with a density of 5.44 g/cm^3 . The largest density in the PbF_2 – PbBr_2 – P_2O_5 based systems is 5.81 g/cm^3 with a T_g of 227°C . A reasonable larger density and a low T_g glass in the PbCl_2 – PbBr_2 – P_2O_5 based systems is sample 42 which has a density of 5.25 g/cm^3 and a T_g of 156°C . The stability indexes of glasses in the two groups of systems do not differ within errors of measurement.

Table 1

Batch composition and analyzed composition of glasses in the PbCl_2 – PbBr_2 – P_2O_5 based systems (mol%) [14]

Glass	Composition (mol%)					
	Pb (± 0.17)	P (± 0.16)	F (± 0.03)	Cl (± 0.24)	Br (± 0.16)	O (± 0.77)
B5 (a)	16.67	14.29	0	23.81	9.52	35.71
(b)	16.1	8.29	0	19.80	8.12	47.69
B6 (a)	16.67	14.28	0	19.06	14.29	35.70
(b)	15.41	7.99	0	21.6	10.04	44.96
B27 (a)	17.48	13.94	0	18.59	13.93	36.06
(b)	16.02	6.23	0	18.95	9.53	49.27
B42 (a)	17.27	13.77	2.41	18.35	13.77	34.43
(b)	16.09	6.3	3.24	19.18	8.6	46.59
B52 (a)	17.37	13.84	1.21	18.48	13.86	35.24
(b)	15.66	6.61	2.98	18.72	10.99	45.04
B67 (a)	14.58	16.41	2.27	8.2	16.40	42.14
(b)	14.5	7.29	3.29	12.42	10.87	51.63

Note: (a) batch composition; (b) analyzed composition.

Table 2

Batch composition and analyzed composition of glasses in the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems (mol%) [15]

Sample	Composition				
	Pb (± 0.19)	F (± 0.14)	P (± 0.17)	Br (± 0.27)	O (± 0.78)
A11 (a)	13.04	13.04	17.40	13.04	43.48
(b)	14.84	1.95	7.83	13.93	61.49
A14 (a)	13.05	4.33	17.40	21.75	43.47
(b)	14.84	1.95	7.83	13.90	61.49
A17 (a)	13.04	0.00	17.40	26.09	43.47
(b)	14.94	0.00	7.38	14.33	63.35
A20 (a)	16.66	14.30	14.29	19.04	35.71
(b)	18.87	3.43	3.35	27.22	47.13
A21 (a)	16.67	9.52	14.28	23.82	35.71
(b)	18.81	3.43	3.35	27.22	47.13
A45 (a)	14.75	12.45	16.59	12.45	43.77
(b)	16.08	4.85	7.69	16.14	55.25
A53 (a)	14.75	4.14	16.60	20.74	43.77
(b)	12.68	1.95	6.53	10.63	68.21
A54 (a)	14.75	0.00	16.59	24.89	43.77
(b)	13.72	0.00	11.41	11.48	63.38

Note: (a) batch composition; (b) analyzed composition.

Table 3

The characteristic temperatures, stability indices and density of some glasses in the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems [11]

Sample	Composition (mol%)				T_g (± 2) ($^\circ\text{C}$)	T_x (± 2) ($^\circ\text{C}$)	H' (± 0.2)	Density (± 0.01) (g/cm^3)
	P_2O_5	PbF_2	PbBr_2	PbO				
A8	40	42	18	0	281	378	0.35	5.59
A11	40	30	30	0	258	376	0.46	5.37
A13	40	18	42	0	247	373	0.51	5.34
A16	40	6	54	0	260	370	0.42	5.28
A48	27	18	45	10	227	281	0.24	5.81
A52	36	18	36	10	210	278	0.32	5.44
A53	36	9	45	10	228	287	0.26	5.40

Note: T_g – glass transition temperature; T_x – onset temperature of crystallization.

Table 4

The characteristic temperatures, stability indices and density of some glasses in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems [12]

Sample	Composition (mol%)					T_g (± 2) ($^\circ\text{C}$)	T_x (± 2) ($^\circ\text{C}$)	H' (± 0.2)	Density (± 0.01) (g/cm^3)
	P_2O_5	PbBr_2	PbCl_2	PbO	PbF_2				
B5	30	20	50	0	0	173	233	0.35	4.84
B6	30	30	40	0	0	146	203	0.39	4.75
B27	28.5	28.5	38	5	0	190	240	0.26	5.46
B42	28.5	28.5	38	0	5	156	230	0.47	5.25
B52	28.5	28.5	38	2.5	2.5	186	244	0.31	5.43
B67	36	36	18	5	5	175	260	0.49	–

Note: T_g – glass transition temperature; T_x – onset temperature of crystallization.

Figs. 6 and 7 show the densities vs. compositions of the glass samples in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ and $\text{PbO--PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ systems, respectively. The figures show that increasing the

amount of heavy components results in larger densities of the samples. Since PbO has the largest density and P_2O_5 has the smallest density among the components, changes in the PbO content and/

Table 5

Water durability data of some glasses in the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems [11]

Sample	Composition				Water durability	Appearance after exposure to air for 180 days
	P_2O_5	PbF_2	PbBr_2	PbO		
A11	40	30	30	0	D	Powder, opaque
A12	40	24	36	0	C	Translucent
A13	40	18	42	0	D	Powder, opaque
A14	40	12	48	0	B	Translucent
A15	40	10	50	0	A	Transparent, no visible change
A38	60	40	0	0	D	Powder, opaque
A48	27	18	45	10	A	Transparent, no visible change
A52	36	18	36	10	A	Transparent, no visible change
A54	36	0	54	10	A	Transparent, no visible change
A55	45	18	27	10	A	Transparent, no visible change
A56	45	27	18	10	A	Transparent, no visible change
A58	54	27	9	10	A	Transparent, no visible change
A142	70	0	0	30	D	Powder, opaque

Note: A: very good; B: fair; C: poor; D: very poor.

Table 6

Water durability data of some glasses in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems [16]

Sample	Composition (mol%)					Dissolution rate in water (mm/day)
	P_2O_5	PbBr_2	PbCl_2	PbO	PbF_2	
B1	40	0	60	0	0	1.0×10^{-4}
B2	40	10	50	0	0	1.3×10^{-4}
B3	40	20	40	0	0	8.1×10^{-5}
B4	40	30	30	0	0	2.1×10^{-4}
B5	30	20	50	0	0	7.9×10^{-5}
B6	30	30	40	0	0	9×10^{-5}
B7	50	30	20	0	0	1.0×10^{-2}
B11	60	30	10	0	0	2.2×10^{-2}
B27	28.5	28.5	38	5	0	5×10^{-5}
B42	28.5	28.5	38	0	5	9.8×10^{-5}
B52	28.5	28.5	38	2.5	2.5	1.1×10^{-4}

or P_2O_5 content will make the largest changes in the densities of the samples.

Tables 5 and 6 list the water durability data of some samples in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ and $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems. Samples in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems have good water durability (the dissolution rate in water is in the order of 10^{-4} mm/day or less) when the P_2O_5 content ≤ 40 mol%. In the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary system containing 40 mol% P_2O_5 , water durable glasses can be obtained only when the PbF_2 content is low (≤ 10 mol%). The addition of 10 mol% PbO into the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary

system improved the water durability of the samples. Glass samples containing 10 mol% PbO in this system remained water durable even when the P_2O_5 content was increased to 54 mol% (Table 5).

4. Discussion

The introduction of PbO and/or PbF_2 into the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ system affects the glass-forming region in three ways. First, it decreases the glass-forming tendency by decreasing the P_2O_5

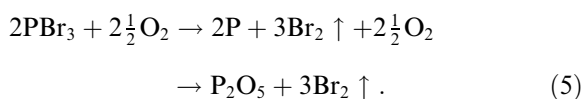
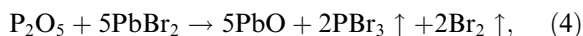
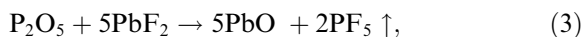
content (the composition shown in the triangle in Figs. 2 and 4 is the ratio of the three components, not the actual contents); second, it increases the glass-forming tendency through the mixed-ion effect; and third, it changes the liquidus temperature of the mixture so that some compositions become difficult to melt at the melting temperatures in this experiment. The overall effect will be a combination of the three. Through comparison with the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary system, the glass-forming regions in both the $\text{PbO--PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ system and the $\text{PbF}_2\text{--PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ system shrink towards smaller PbCl_2 content and extend towards larger PbBr_2 content. The mixed-ion effect makes the glass-forming region larger, but since the melting temperatures of both PbO and PbF_2 are higher than those of the other components, the addition of these components makes the compositions with large PbCl_2 content melt at higher temperatures not used in this research. Thus, the glass-forming regions can only extend PbBr_2 .

The difference in the T_g and densities of samples in the two systems is mainly caused by two factors. The first is the lower melting point and smaller density of PbCl_2 than those of PbF_2 , and the second is the different melting temperatures. As noted above, the melting temperature of the glasses in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems is lower than that in the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems. Therefore, the decomposition of $\text{NH}_4\text{H}_2\text{PO}_4$ in the former systems is less complete than that in the latter systems and the glasses in the former systems contain hydroxyl (OH) and ammonium [13]. Since OH in glasses will disrupt the network connectivity and make diffusion easier, it decreases the glass transition temperature, the density and the glass-forming tendency of the samples. The glass melting temperature also affects the final composition of the glasses. Higher temperature increases evaporation of scrap components and results in greater difference in the actual composition and the batch composition. Thus, samples with the same batch composition will have different glass-forming tendencies if melted at different temperatures. When melted at a higher temperature (530–540°C), the glass-forming tendency in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ ternary system increases and the

glass-forming region is extended (Fig. 5) [13]. The glass transition temperature and the densities of the samples are increased. We assume the reason for these change to be the different amounts of hydroxyl and ammonium in the samples.

In most cases, the density of a glass is dependent upon the densities of the components in the glass. Increasing the percentage of heavy components will increase the density of the glasses. In Fig. 6, however, the densities of the samples do not decrease with the increase of the ratio of $\text{PbCl}_2/(\text{PbCl}_2 + \text{PbBr}_2)$ when the ratio is less than 0.8, although the density of PbCl_2 (5.85 g/cm³) is less than that of PbBr_2 (6.66 g/cm³). This difference is attributed to the differences in the vaporization rates of PbBr_2 and PbCl_2 . Since the melting point of PbBr_2 (373°C) is less than that of PbCl_2 (501°C), the vaporization rate of PbBr_2 is greater than that of PbCl_2 . Thus, for glasses containing more PbBr_2 , the total amount of vaporized PbBr_2 and PbCl_2 is larger and this loss causes the actual P_2O_5 content to be relatively larger and the density of the sample to be relatively smaller.

The evaporation of the components during glass melting is complicated. Through an analysis, the loss of the components in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems when melted at 450–470°C is assumed to be caused mainly through the evaporation of PbBr_2 , P_2O_5 and PbCl_2 . The reaction between P_2O_5 and the lead halides to form phosphorus halides, if any, is negligible [14]. For glasses in the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems melted at 600°C, however, the following reactions also contribute to the loss of the components in addition to the evaporation of PbBr_2 , P_2O_5 and PbCl_2 [15].



As a result of these reactions, phosphorus, bromine and fluorine are lost and oxygen comes into the melt from the air.

5. Conclusions

1. Glasses can be formed in the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems containing 0–5 mol% of PbO and/or PbF_2 and in the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ based systems containing ≤ 15 mol% PbO . The dissolution rate in water at room temperature of most heavy glasses in the systems is in the order of 10^{-5} mm/day. The water durable glasses in the systems have densities in the range 4.5–5.81 g/cm³ and have glass transition temperatures in the range 146–281°C.
2. Introduction of PbO and/or PbF_2 into the $\text{PbCl}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ system increases the density and glass transition temperature of the glasses, while introduction of 10 mol% PbO into the $\text{PbF}_2\text{--PbBr}_2\text{--P}_2\text{O}_5$ system decreases the glass transition temperature and increases the density. Some components are lost during the melting process and this makes the analyzed composition of a glass different from the batch composition consequently. Melting temperature has an effect on the glass-forming tendency and the properties of the glasses.

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